MILITARY SPECIFICATION

KEY

1. SCOPE

1.1 This specification covers requirements, quality assurance provisons, and the preparation for delivery criteria for one type of key used on XM455 projectile.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATION
MILITARY
MIL-P-116 Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Table for Inspection by Attributes
MIL-STD-1167	Ammunition Data Cards
MIL-STD-1168	Lot Numbering of Ammunition
MIL-STD-1169(MU)	Packaging, Packing and Marking for Shipments of Inert Ammunition

DRAWINGS

ORDNANCE CORPS

8808362	Packing and Marking for Storage and Shipments
8883536	Bag, Packing, for repair Parts
10522823	Key
EL-10522823	Equipment List

FSC 1135

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The key shall comply with all requirements specified on the drawing (dwg) 10522823, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 Workmanship.

3.3.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.3.2 Burrs. Parts shall be free from burrs.

3.3.3 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.3.4 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable

4.3 Quality conformance inspection.

4.3.1 Lot formation.

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4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table I and Table II), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table I
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

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Table II
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Key (see dwg 10522823).

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major		
101	Outside curvature (1.882)	Gage
102	Inside curvature of top lugs (1.82)	Gage
103	Curvature of bottom lugs (1.25)	Gage
104	Length of top lugs (.251)	Gage
105	Total width (1.520)	Gage
106	Diameter of counterbore (.29)	Gage
107	Depth of counterbore, min (.19)	Gage
108	Total thickness top (.480)	Gage
109	Distance from bottom of groove to right face (.38)	Gage
110	Location of hole	Gage
111	Total thickness bottom (.54)	Gage
112	Width of bottom lugs (.12)	Gage
113	Material (see 3.1)	Visual
114	Objective evidence (see 6.2) for protective finish in accordance with dwg 10522823	Visual
Minor		
201	Surface finish	Visual
202	Workmanship (see 3.3)	Visual

4.3.2.1.2 Preparation for delivery (see 5.1).

Categories	Characteristics	Method of Inspection
Critical: None defined		

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Categories	Characteristics	Method of Inspection
Major		
101	Preservation, packaging, packing and marking in accordance with MIL-STD-1169(MU)	Visual-Manual

Minor: None defined

4.3.2.1.3 Bag, packing for repair parts (see dwg 8883536 covering dwg part number 8883536-2) and 5.2.

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major		
101	Material (see 3.1)	Visual
102	Length before packing, min	*Measure
103	Width before packing, min	*Measure
104	Properly packed	Visual-Manual
105	Proper heat seal	Visual-Manual
106	No crack or hole in bag	Visual
107	Objective evidence (see 6.2) for heat seal test in accordance with MIL-P-116	Visual
*Measure with suitable scale		
Minor		
201	Marking	Visual
202	Workmanship (see 3.3)	Visual-Manual

4.3.2.1.4 Packing and marking for storage and shipments (see dwg 8808362 and 5.2).

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major		
101	Gross load weight, max	*Measure
102	Contents tight in package	Manual
103	Properly packed	Visual-Manual
104	Material (see 3.1)	Visual
*Determine weight with suitable scale		

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Categories	Characteristics	Method of Inspection
Minor		
201	Marking	Visual
202	Workmanship (see 3.3)	Visual-Manual

4.3.3 Testing. Not applicable

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (See EL-10522823). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).

b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. Not applicable

5. PREPARATION FOR DELIVERY

5.1 Production items.

5.1.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1.1 Level A. The key shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.1.2 Level C. The key shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Packing.

5.1.2.1 Levels A and C. The key shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

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5.1.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

5.2 Repair parts.

5.2.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.2.1.1 Level A. The key shall be packaged in accordance with dwg 8883536 and 8808362 using the specified moisture barrier.

5.2.1.2 Level C. The key shall be packaged in accordance with dwg 8808362 except that the moisture barrier may be omitted.

5.2.2 Packing.

5.2.2.1 Level A and C. The key shall be packed in accordance with dwg 8883536 and 8808362.

5.2.3 Marking. Marking shall be as specified in accordance with dwg 8883536 and 8808362 as applicable.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Production or Repair Parts.
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on DI-E-1118 or specification when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-N-I.

Custodian:
Army - MU

Preparing activity:
Army - MU

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